Practice Exam 1 (F25)

Started: Sep 25 at 6:08pm

Quiz Instructions

Question 1 1 pts

Which file types are mostly or entirely in machine language binary and unintelligible to humans? Choose U for unintelligible and I for intelligible.

1. C file [Select]
2. Assembly file [Select]
3. Object file [Select]
4. Executable file [Select]

Question 2 1 pts

The tool that translates high-level language to assembly is called [Select] .

The tool that combines object files to create one executable is called [Select] .

The tool that translates assembly to machine code is called [Select] .

Question 3 1 pts

Which ones are acceptable ways to declare a pointer?

- int *a; [Select]
- int &a; [Select]
- int a*; [Select]

- int a&; [Select]

Question 4 1 pts

What is the value in the last variable in each of the statements below (the int # variable in the last line)?

1. int x = 5;
2. int* y = &x;

[Select]

1. int x = 5;
2. int* y = &x;
3. int z = *y;

[Select]

1. int x = 5;
2. int* y = &x;
3. y = y + 1;
4. int z = *y;

[Select]

Question 5 1 pts

```c
typedef struct {
    int x, y, z;
    char c;
} bunch_t;


void WorkwithBunch(bunch_t inputBunch)


// later inside the main:
   bunch_t myBunch = {1, 2, 3, '7'};
   bunch_t *bunchPtr = &myBunch;

// How should we call WorkwithBunch?
   WorkwithBunch(bunchPtr); //  option 1
   WorkwithBunch(*bunchPtr); // option 2
   WorkwithBunch(&bunchPtr); // option 3
```

Consider the above snippet. What option is the correct way to call the WorkwithBunch function?

- ○ Option 1
- ○ Option 2
- ○ Option 3
- ○ None

Question 6 1 pts

1. unsigned int* ptr = (unsigned int*)0x4000C000;
2. ptr = ptr + 2;

What is the value of ptr after the addition? Answer in the format 0x########, capitalize all letters.

Question 7 1 pts

How many bits in a signed character? [Select]

How many bits in an integer (MSP432)? [Select]

Question 8 1 pts

Which one is bit 23 mask (BIT23)?

- (1<<22)
- (1<<23)
- (1<<24)

Question 9 1 pts

A single-bit bitmask is a bitstring where all the bits are 0 except for a single bit, which is 1. If that single bit is located at index i, we call the bitmask BITi. For example, BIT0 is equal to 1 and BIT1 is equal to 2 and BIT2 is equal to 4, and so on. If we intend to reset bits 10 and 20 of an integer, and we like to create a macro mask, what should go in the blank space below?. If we intend to reset bits 10 and 20 of an integer, and we like to create a macro mask, what should go in the blank space below?

```
#define NEW_MASK ....
x = x & ~NEW_MASK;
```

- (BIT10 | BIT20)
- BIT10 | BIT20
- BIT10 & BIT20
- ~(BIT10 | BIT20)

Question 10 1 pts

On the Launchpad board, Button 2 (S2) is connected to Port [ ] , Pin [ ] and the red LED 2 (LED2_RED) is connected to Port [ ] , Pin [ ] .

Please only write the numeric value itself.

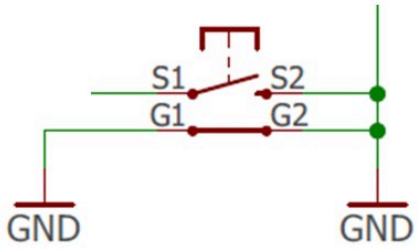
Question 11 1 pts

Consider a Schmitt trigger with two thresholds at 1.4V and 1.6V. Also, assume the logic 0 is 0V, and logic 1 is 3V.

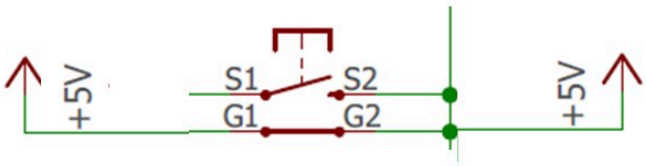
1. What is the output of the Schmitt trigger when the input is 0.2V? [Select]
2. What is the output of the Schmitt trigger when the input is 1.55V? [Select]
3. What is the output of the Schmitt trigger when the input is 2.33V? [Select]

Question 12 1 pts

Consider the two below setups for a pushbutton. In order for them to work properly which statement should be correct?



Configuration (A)



Configuration (B)

1. Configuration A [Select] .
2. Configuration B [Select] .

Question 13 1 pts

In MSP432,

- if we wish to configure a pin as an input with a pull-down resistor, DIR register [Select] , REN register [Select] , and OUT register [Select] .

Question 14 1 pts

What is the address of the P9DIR register?

What is the address of the P10SELC register?

Your answer should be in the format 0x########, capitalize all letters.

Question 15 1 pts

You need to study one or more of the technical documents we discussed in class to answer this question.

The red LED on the booster is connected to which BoosterPack pin?

- J4.39
- J4.41
- P2.6

Question 16 1 pts

What topic covered in class that places a layer between the software and hardware allows us to easily port high-level code from one processor to another?

Quiz saved at 8:31pm Submit Quiz